

# Contribution Journal

Group Members Names:

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Class: 6263-ITAI-1371 Intro to Machine Learning-S10-12321

## Edwin Alvarado:

For this lab, I wrote the report and analysis writeup. I pulled together the work each of us did into one coherent document, connecting the dataset generation, the three polynomial models, and the learning curves into a single narrative. I explained why Degree 1 underfits from high bias, why Degree 4 gives the best fit, and why Degree 15 overfits from high variance, then tied that back to what the learning curves showed about how each model generalizes. I also cleaned up the wording, ordered the sections logically, and made sure the analysis and conclusions matched the results shown in the code and visualizations.

## Kala Hayes

For this lab, I helped write and run the code used for the differential polynomial models. I also assisted with debugging by fixing indentation and syntax errors, resolving coding issues, and writing code to ensure that the models and visualizations were generated correctly. As well uploaded the documents to GitHub.

## Anashrah Adil

During this lab, I generated and visualized a synthetic sine wave dataset with random noise using Python's numpy and matplotlib libraries. I then built and trained three polynomial regression models of varying complexities Degrees 1, 4, and 15 using scikit-learn to demonstrate model performance. Finally, I visualized and analyzed the behavior of each model, identifying Degree 1 as underfitting due to high bias, Degree 4 as providing an optimal fit, and Degree 15 as overfitting due to high variance, thereby effectively illustrating the bias-variance tradeoff.

## Zain Ahmed:

For this lab, I worked on the learning curves section. I built the underfit and overfit pipelines and ran them through scikit-learn's `learning_curve` function with 5-fold cross-validation, then plotted how the training score and cross-validation score moved as the training set size grew. For the

underfit model, both curves stayed low and close together; for the overfit model, the training score stayed high while the cross-validation score lagged behind, leaving a wide gap. I used that gap to write up the knowledge check answers, explaining what each curve shape means about how the model is generalizing.